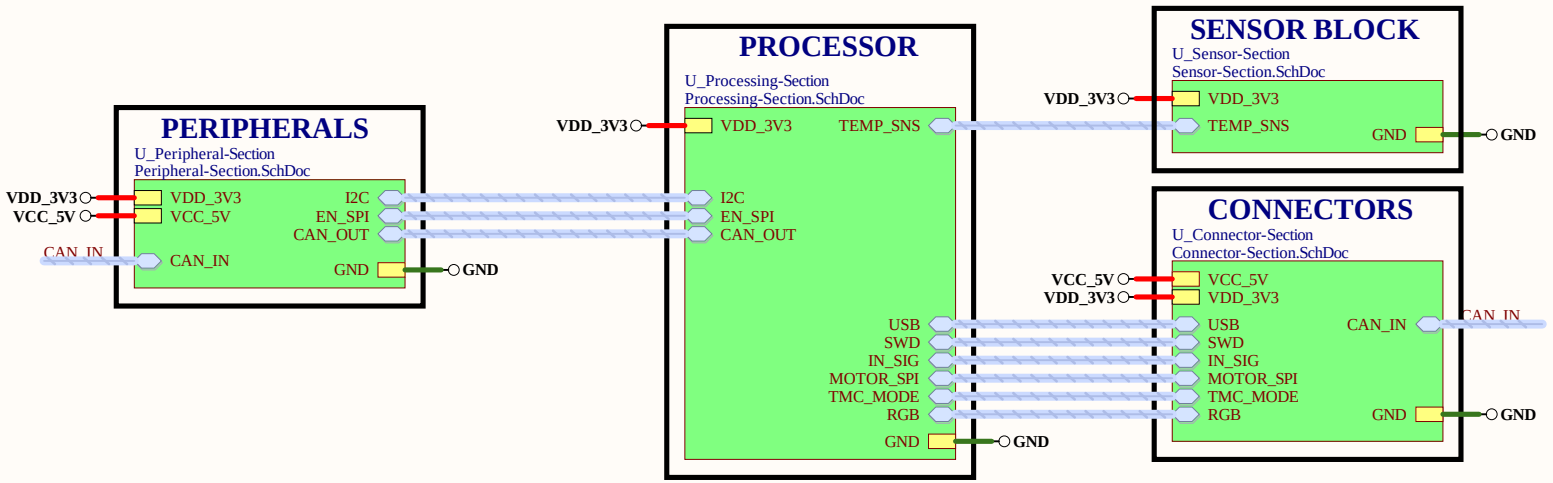
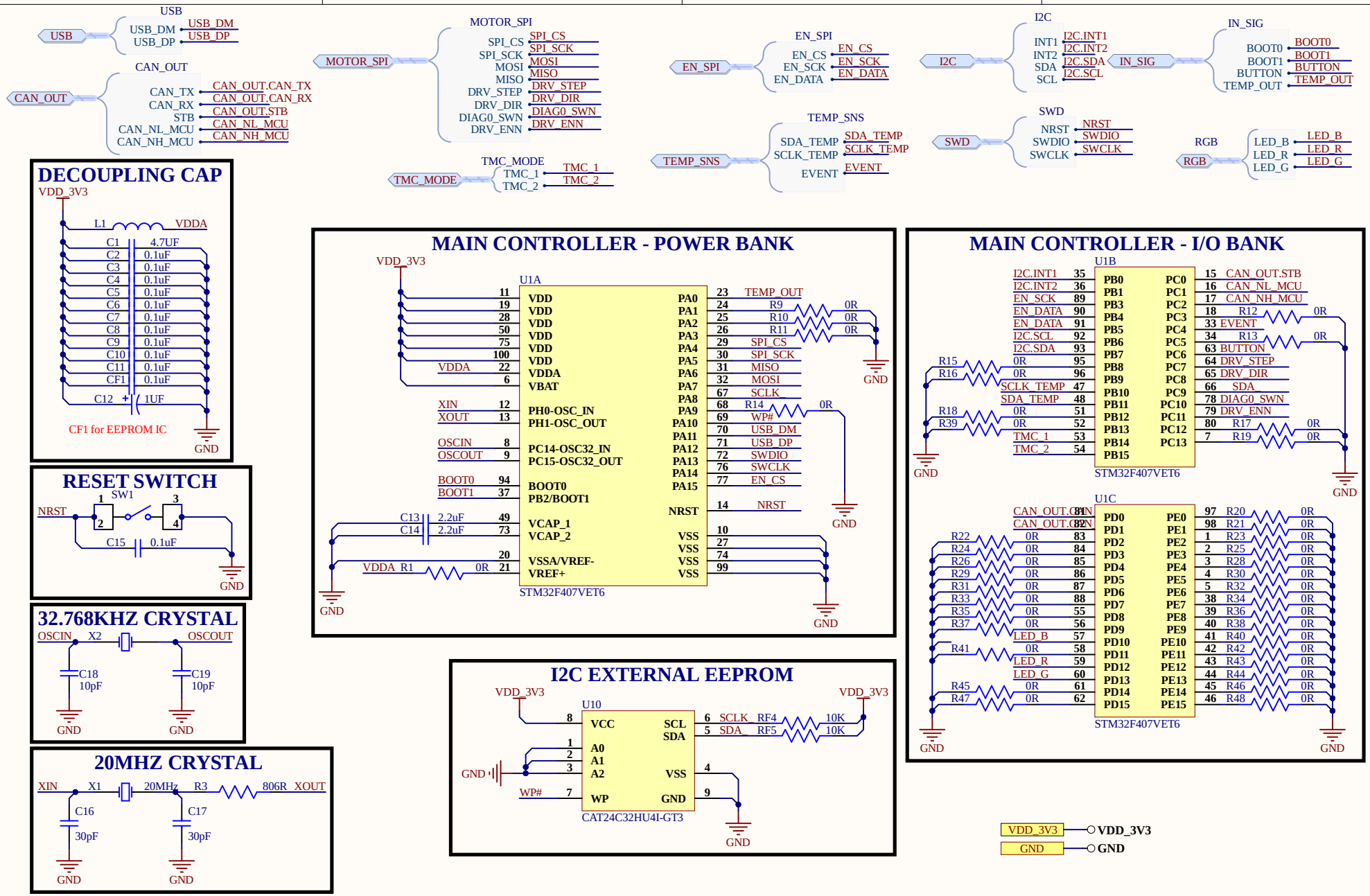
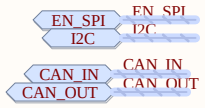
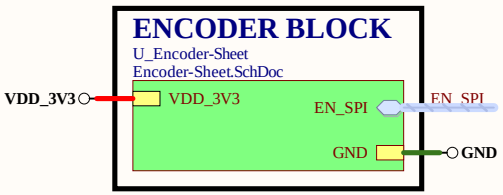
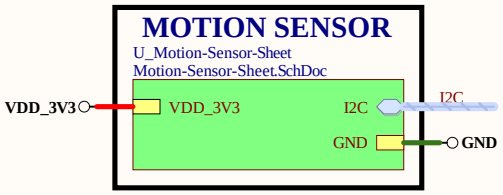
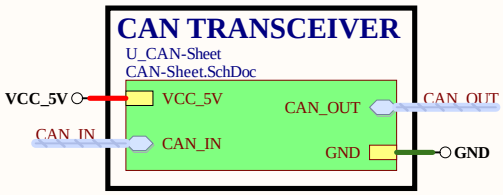

PERIPHERALS
U_Peripheral-Section
Peripheral-Section.SchDoc
VDD_3V3
VCC_5V
CAN_IN
VDD_3V3
VCC_5V
CAN_IN
I2C
EN_SPI
CAN_OUT
GND
GND
PROCESSOR
U_Processing-Section
Processing-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
I2C
EN_SPI
CAN_OUT
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
GND
GND
SENSOR BLOCK
U_Sensor-Section
Sensor-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VCC_5V
VDD_3V3
VCC_5V
VDD_3V3
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
CAN_IN
CAN_IN
GND
GND

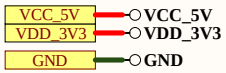

USB
USB
USB_DM
USB_DP
USB_DM
USB_DP
CAN_OUT
CAN_OUT
CAN_TX
CAN_RX
STB
CAN_NL_MCU
CAN_NH_MCU
CAN_OUT.CAN_TX
CAN_OUT.CAN_RX
CAN_OUT.STB
CAN_NL_MCU
CAN_NH_MCU
MOTOR_SPI
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TMC_MODE
TMC_MODE
TMC_1
TMC_2
EN_SPI
EN_SPI
EN_CS
EN_SCK
EN_DATA
I2C
I2C
INT1
INT2
SDA
SCL
I2C.INT1
I2C.INT2
I2C.SDA
I2C.SCL
IN_SIG
IN_SIG
BOOT0
BOOT1
BUTTON
TEMP_OUT
TEMP_SNS
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SWD
SWD
NRST
SWDIO
SWCLK
RGB
RGB
LED_B
LED_R
LED_G
DECOUPLING CAP
VDD_3V3
L1
VDDA
C1 4.7UF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
CF1 0.1uF
C12 1UF
CF1 for EEPROM IC
GND
MAIN CONTROLLER - POWER BANK
VDD_3V3
U1A
VDD 11
VDD 19
VDD 28
VDD 50
VDD 75
VDD 100
VDDA 22
VBAT 6
PH0-OSC_IN 12
PH1-OSC_OUT 13
PC14-OSC32_IN 8
PC15-OSC32_OUT 9
BOOT0 94
PB2/BOOT1 37
VCAP_1 49
VCAP_2 73
VSSA/VREF- 20
VREF+ 21
PA0 23 TEMP_OUT
PA1 24 R9 0R
PA2 25 R10 0R
PA3 26 R11 0R
PA4 29 SPI_CS
PA5 30 SPI_SCK
PA6 31 MISO
PA7 32 MOSI
PA8 67 SCLK
PA9 68 R14 0R
PA10 69 WP#
PA11 70 USB_DM
PA12 71 USB_DP
PA13 72 SWDIO
PA14 76 SWCLK
PA15 77 EN_CS
NRST 14 NRST
VSS 10
VSS 27
VSS 74
VSS 99
XIN 12
XOUT 13
OSCIN 8
OSCOUT 9
C13 2.2uF
C14 2.2uF
VDDA_R1 0R
STM32F407VET6
MAIN CONTROLLER - I/O BANK
U1B
I2C.INT1 35
I2C.INT2 36
EN_SCK 89
EN_DATA 90
EN_DATA 91
I2C.SCL 92
I2C.SDA 93
SCLK_TEMP 47
SDA_TEMP 48
TMC_1 53
TMC_2 54
PB0 PC0
PB1 PC1
PB3 PC2
PB4 PC3
PB5 PC4
PB6 PC5
PB7 PC6
PB8 PC7
PB9 PC8
PB10 PC9
PB11 PC10
PB12 PC11
PB13 PC12
PB14 PC13
PB15
15 CAN_OUT.STB
16 CAN_NL_MCU
17 CAN_NH_MCU
18 R12 0R
33 EVENT
34 R13 0R
63 BUTTON
64 DRV_STEP
65 DRV_DIR
66 SDA
78 DIAG0_SWN
79 DRV_ENN
80 R17 0R
7 R19 0R
R15 0R
R16 0R
R18 0R
R39 0R
STM32F407VET6
U1C
PD0 PE0
PD1 PE1
PD2 PE2
PD3 PE3
PD4 PE4
PD5 PE5
PD6 PE6
PD7 PE7
PD8 PE8
PD9 PE9
PD10 PE10
PD11 PE11
PD12 PE12
PD13 PE13
PD14 PE14
PD15 PE15
97 R20 0R
98 R21 0R
1 R23 0R
2 R25 0R
3 R28 0R
4 R30 0R
5 R32 0R
38 R34 0R
39 R36 0R
40 R38 0R
41 R40 0R
42 R42 0R
43 R43 0R
44 R44 0R
45 R46 0R
46 R48 0R
R22 0R
R24 0R
R26 0R
R29 0R
R31 0R
R33 0R
R35 0R
R37 0R
R41 0R
R45 0R
R47 0R
LED_B 57
LED_R 59
LED_G 60
STM32F407VET6
RESET SWITCH
NRST
SW1
C15 0.1uF
32.768KHZ CRYSTAL
OSCIN
X2
OSCOUT
C18 10pF
C19 10pF
20MHZ CRYSTAL
XIN
X1
20MHz
R3 806R
XOUT
C16 30pF
C17 30pF
I2C EXTERNAL EEPROM
VDD_3V3
U10
VCC 8
A0 1
A1 2
A2 3
WP# 7
WP
SCL 6 SCLK_RF4 10K
SDA 5 SDA_RF5 10K
VSS 4
GND 9
CAT24C32HU4I-GT3
VDD_3V3
VDD_3V3
GND
GND


EN_SPI
I2C
EN_SPI
I2C
CAN_IN
CAN_OUT
CAN_IN
CAN_OUT

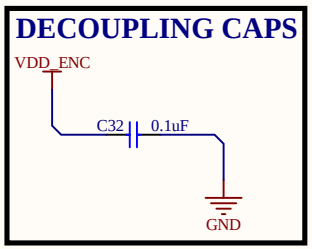

ENCODER BLOCK
U_Encoder-Sheet
Encoder-Sheet.SchDoc
VDD_3V3
VDD_3V3
EN_SPI
EN_SPI
GND
GND
GND

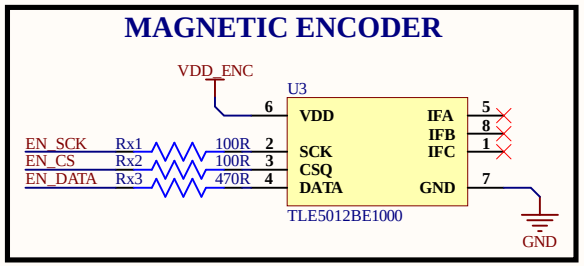

MOTION SENSOR
U_Motion-Sensor-Sheet
Motion-Sensor-Sheet.SchDoc
VDD_3V3
VDD_3V3
I2C
I2C
GND
GND
GND

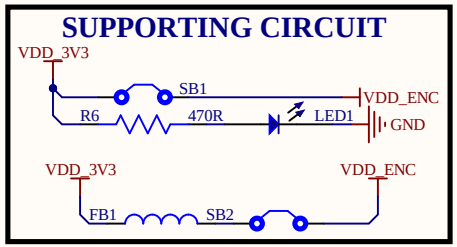

CAN TRANSCEIVER
U_CAN-Sheet
CAN-Sheet.SchDoc
VCC_5V
VCC_5V
CAN_OUT
CAN_OUT
CAN_IN
CAN_IN
GND
GND
GND

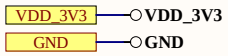

VCC_5V
VDD_3V3
GND
VCC_5V
VDD_3V3
GND

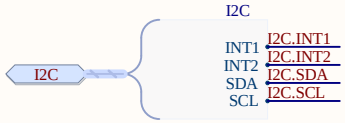

EN_SPI
EN_SPI
EN_CS
EN_SCK
EN_DATA
EN_CS
EN_SCK
EN_DATA

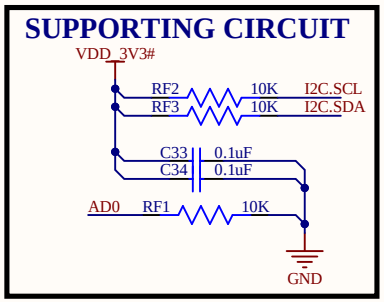

DECOUPLING CAPS
VDD_ENC
C32
0.1uF
GND

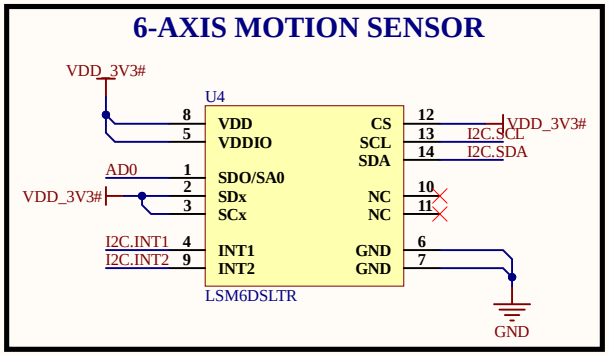

MAGNETIC ENCODER
VDD_ENC
U3
6
VDD
IFA
5
IFB
8
IFC
1
EN_SCK
Rx1
100R
2
SCK
EN_CS
Rx2
100R
3
CSQ
EN_DATA
Rx3
470R
4
DATA
GND
7
TLE5012BE1000
GND

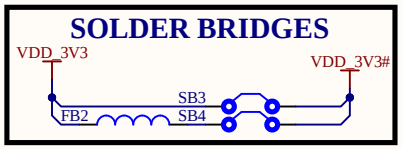

SUPPORTING CIRCUIT
VDD_3V3
SB1
VDD_ENC
R6
470R
LED1
GND
VDD_3V3
VDD_ENC
FB1
SB2

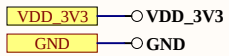

VDD_3V3
VDD_3V3
GND
GND

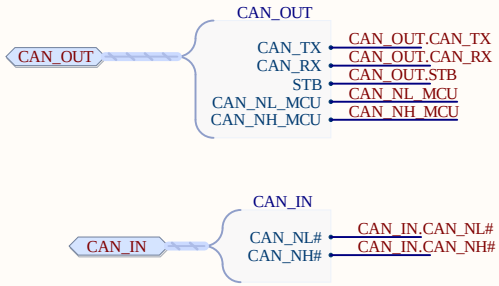

I2C
I2C
INT1
INT2
SDA
SCL
I2C.INT1
I2C.INT2
I2C.SDA
I2C.SCL


SUPPORTING CIRCUIT
VDD_3V3#
RF2
10K
I2C.SCL
RF3
10K
I2C.SDA
C33
0.1uF
C34
0.1uF
AD0
RF1
10K
GND

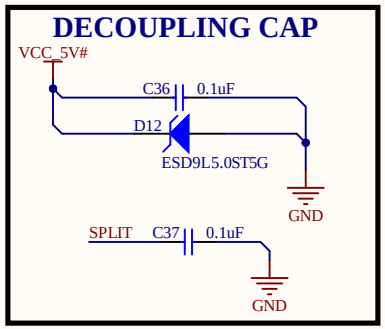

6-AXIS MOTION SENSOR
VDD_3V3#
U4
8
VDD
5
VDDIO
AD0
1
SDO/SA0
VDD_3V3#
2
SDx
3
SCx
I2C.INT1
4
INT1
I2C.INT2
9
INT2
CS
12
SCL
13
SDA
14
NC
10
NC
11
GND
6
GND
7
VDD_3V3#
I2C.SCL
I2C.SDA
LSM6DSLTR
GND

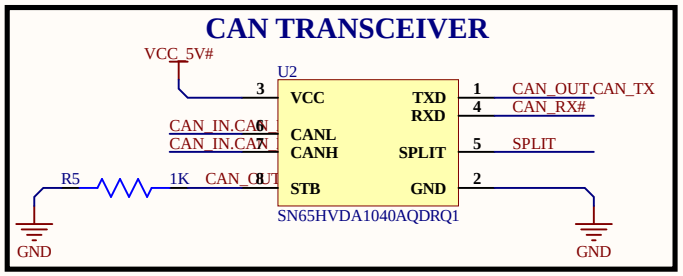

SOLDER BRIDGES
VDD_3V3
VDD_3V3#
SB3
FB2
SB4

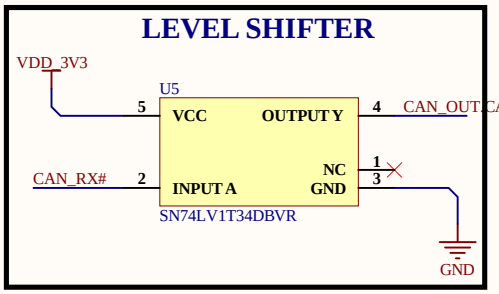

VDD_3V3
VDD_3V3
GND
GND

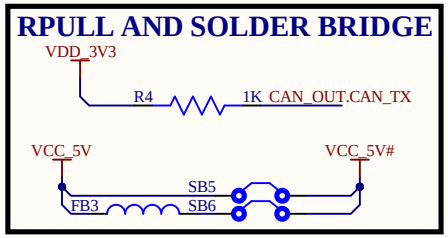


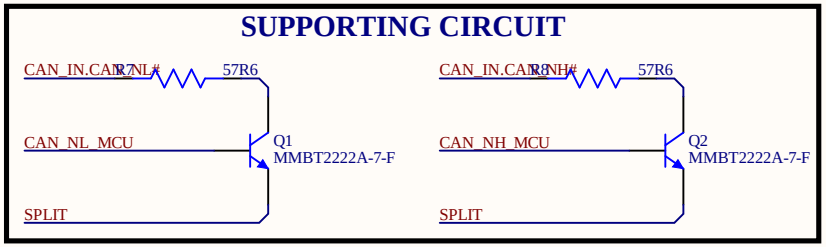

**Connections for CAN are w.r.t Controller Rx and Tx**

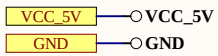


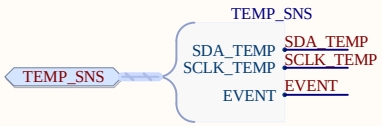



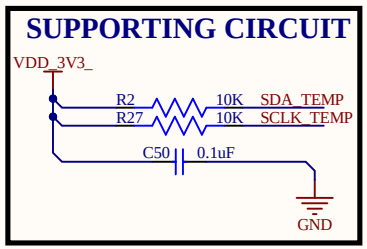



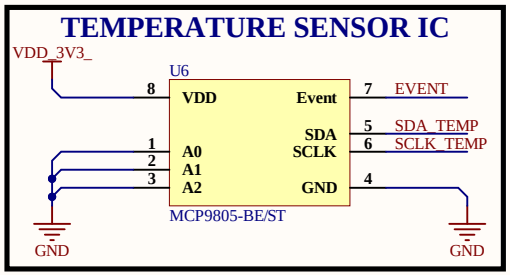



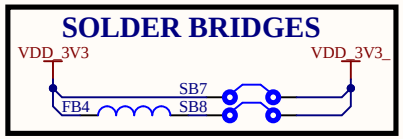



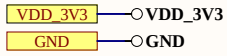



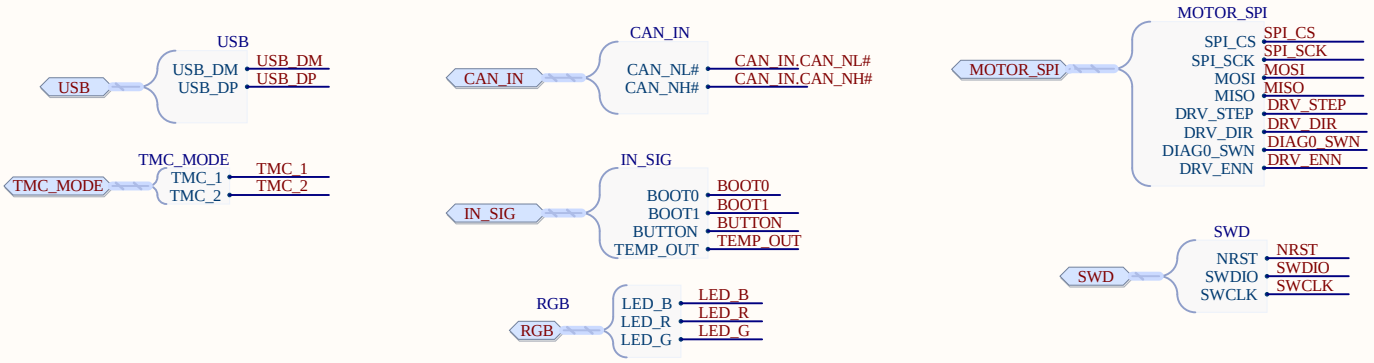

TEMP_SNS
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SDA_TEMP
SCLK_TEMP
EVENT


SUPPORTING CIRCUIT
VDD_3V3_
R2
10K
SDA_TEMP
R27
10K
SCLK_TEMP
C50
0.1uF
GND

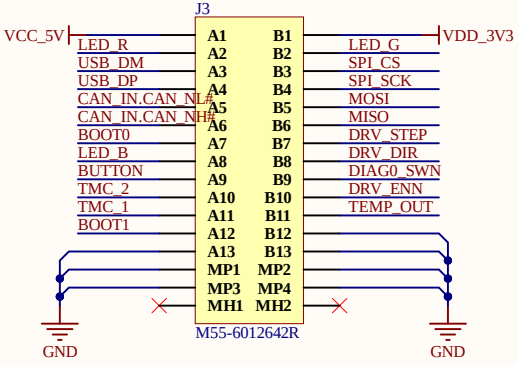

TEMPERATURE SENSOR IC
VDD_3V3_
U6
8
VDD
Event
7
EVENT
SDA
5
SDA_TEMP
1
A0
SCLK
6
SCLK_TEMP
2
A1
3
A2
GND
4
MCP9805-BE/ST
GND
GND


SOLDER BRIDGES
VDD_3V3
VDD_3V3_
SB7
FB4
SB8

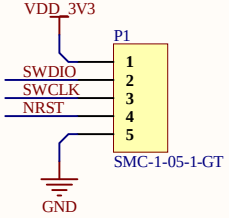

VDD_3V3
VDD_3V3
GND
GND



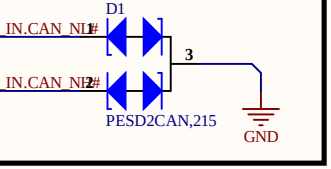

## BOARD TO BOARD CONNECTOR

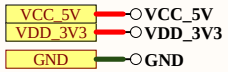


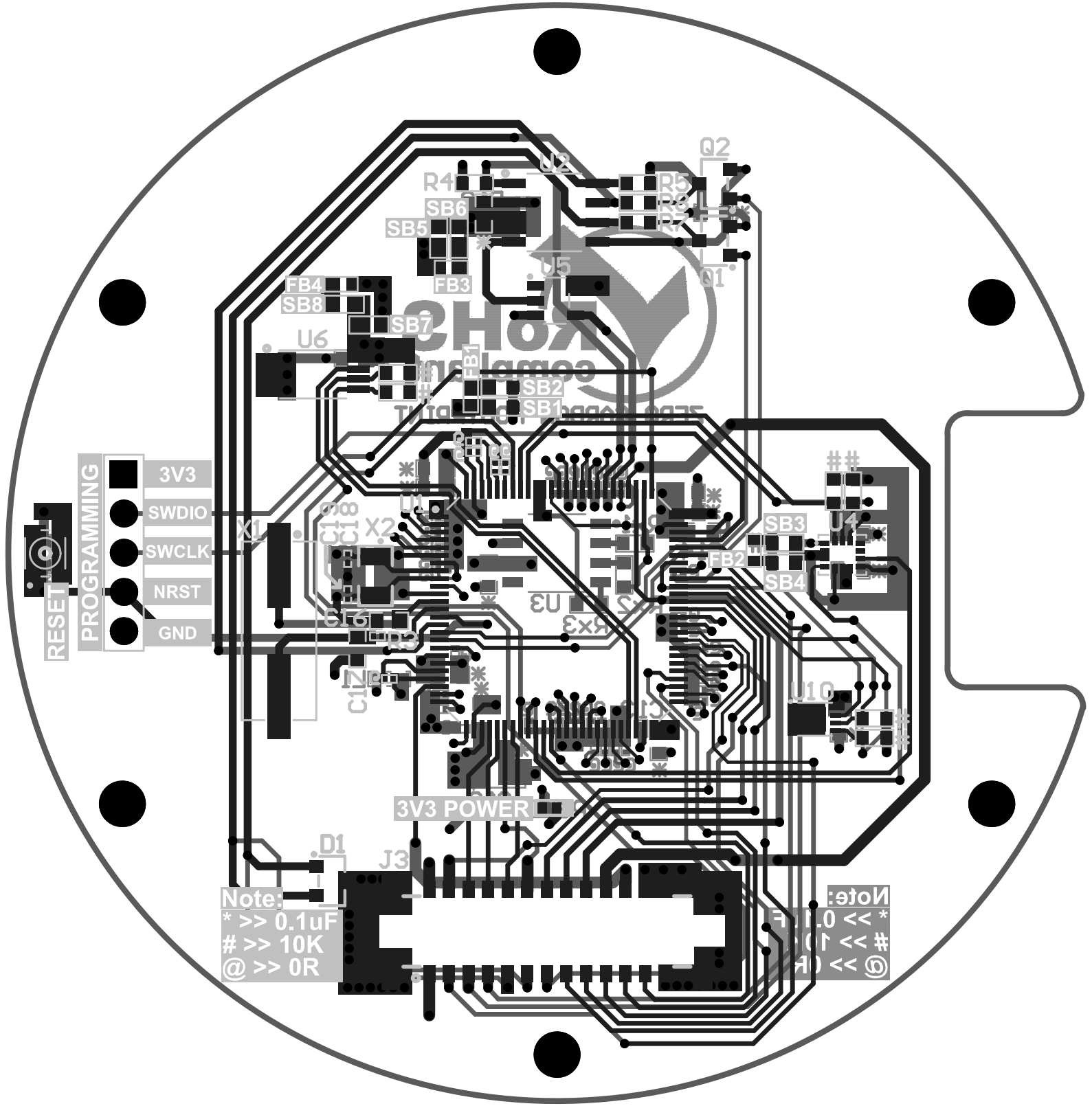

## PROGRAMMING HEADER




## TVS PROTECTION DIODES




3V3
SWDIO
SWCLK
NRST
GND
PROGRAMMING
RESET
3V3 POWER
Note:
* >> 0.1uF
# >> 10K
@ >> 0R